Green Roofs Passively Compensate for Urban Green Space Deficits*

A spatial analysis of Toronto's green roof by-law

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From 2010 to 2025 in Toronto, many newly developed structures were required to install green roofs. Using up-to-date data from Open Data Toronto, we found that, aside from simply expanding total vegetated area, Toronto's green roof by-law helped distribute green surfaces more evenly across the city, allowing for a more efficient mitigation against high rates of urban heating.

1 Introduction

Urban landscapes are often warmer, on average, than surrounding rural regions. This phenomenon (known as the Urban Heat Island Effect, and hereafter referred to as the UHI) is largely driven by the lack of vegetated surfaces (Yang et al. 2016). The primary materials that make up the surfaces of.urban landscapes (e.g. concrete, tar, asphalt and gravel) absorb more solar radiation than landscapes that have not been as developed, such as forests, fields, or farms. Owing to the UHI, among other factors, climate-change-induced global warming is projected to disproportionately affect urban centers (IPCC 2023). In particular, cities are projected to have more intense and longer lasting heat waves (IPCC 2023).

Faced with these unique risks, many municipalities around the world (e.g. Dublin, Ireland and Berlin, Germany) have adopted green roof by-laws (hereafter referred to as GRBLs) to mitigate against the UHI (Basu et al. 2021; Ngan, n.d.). Green roofs are defined by Berardi et al. (Berardi et al. 2014) as "roofs with plants in their final layer", and offer many building-level

*Code and data are available at: https://github.com/bennypk1/starter_folder.

and city-level services including stormwater management and increased public access to nearby green spaces (Berardi et al. 2014). Most importantly for this report, green roofs decrease the overall solar absorbance of an urban area by increasing total vegetated area, with albedos (a measure of surface reflectivity) typically ranging from 0.7 - 0.85, as compared to standard roofing materials (0.1 - 0.2 ; Berardi et al. 2014). This difference has been shown to have cooling effects at local and neighborhood scales (Aram et al. 2019).

Toronto is the first city in North America to have had a By-law requiring the construction of green roofs (City of Toronto 2026a). From January 2010 to November 2025, all new developments and extensions with a floor area in excess of 2000 m^2 were required to construct a green roof, defined by the City as "an extension of [a roof], built on top of a human-made structure, that allows vegetation to grow in a growing medium" (City of Toronto 2026a). Complementing their GRBL, the City also implemented an Eco-Roof Incentive Program (referred throughout as ERIP) to support the voluntary installation of green roofs on homes and buildings (City of Toronto 2026a).

Recent research has found that, while overall area is the most important factor in determining a green roof's ability to offset the UHI, the relative spatial arrangement of green spaces in a city matters as well. For example, Ossola et al. (Ossola et al. 2021) found that the cooling effects of green surfaces in highly urban areas are greater at land unit than in suburbs. In this report, we sought to assess whether or not Toronto's GRBL was effective in homogenizing the distribution of vegetated area in the city by increasing green space coverage in highly urbanized areas of the city. We leveraged data from Open Data Toronto to (1) document the spatial distribution of current ground-level "green spaces" (e.g. parks, etc.), and (2) document the spatial distribution of green roofs, and (3) analyze whether or not green roofs "fill the gaps" missed on the ground. We find that Toronto's green roofs and ground-level green spaces are in complimentary spatial distribution, supporting the claim that green roofs are an effective countermeasure to the UHI.

The remainder of this report is structured as follows. Section 2 provides an overview of the datasets used, as well as summaries of variables relevant for analysis. Section 3 details our findings, and we discuss the broader impact of our results and its limitations in Section 4 and Section 5.

2 Data

Two datasets were used in this report, both sourced from the City of Toronto's Open Data website (City of Toronto 2026b), accessed on May 11, 2026. These are: "Building Permits - Green Roofs" (City of Toronto 2026a), and "Green Spaces" (City of Toronto 2026c), updated weekly and monthly respectively. All data acquisition, cleaning, tabulation, and visualization tasks were conducted within the R Statistical Software ((R Core Team 2025)). Notable packages used in this report include `opendatatoronto` (Gelfand and Toronto 2025),`tidyverse` (Wickham and RStudio 2023), `canadianmaps` (Cayen 2024), `sf` (Pebesma et al. 2026), and `geojsonsf` (Cooley and Teucher 2025).

2.1 Building Permits - Green Roofs

The cleaned "Building Permits - Green Roofs" dataset consists of 1234 building permits related to Toronto's GRBL (n=1193) and ERIP (n=41). Both the GRBL and ERIP were adopted in 2009 by Toronto City Council, and took effect on January 31, 2010. The GRBL was eventually repealed in November 2025, making green roofs voluntary for private developers (City of Toronto 2026a). Permit information of all structures for which the GRBL or ERIP has applied are made public and updated weekly on the City's Open Data website (City of Toronto 2026b), accessed for this report on May 6, 2026. The raw dataset also contained an additional 15 definitively cancelled projects and 5 duplicate entries, which were removed during cleaning.

Each permit is assigned a Permit Number, which consists of the last two digits of its application year (e.g. 11 for 2011), followed by a 6-digit IBMS-generated sequence. After issuance, permit applications may be revised, revisions that may themselves qualify for GRBL or ERIP. Such revisions are recorded using the same Permit Number, and distinguished from the original permit application by an integer value (called the Revision Number). Therefore, identifying truly unique green roofs requires specifying both a Permit and Revision Number. Permit-revision pairs are each assigned types (e.g. "New Building", "Small Residential Projects"), a current status (e.g. "Closed", "Revision Issued"), and a written description of the proposed project. Dates of permit application, issuance, and completion are also provided where applicable. In addition, the full address, forward sortation area (FSA ; i.e. the first three digits of the postal code), type (e.g. "University", "SFD – Detached"), and green roof area (in m^2) of the permit's associated structure is provided.

For this analysis, our variables of interest from this dataset are FSA and green roof area. Table 1 demonstrates what the dataset looked like after cleaning.

Table 1: First 4 Rows of Cleaned Green Roof Data

ID	G_AREA	FSA
03 141785_1	18.36	M8Y
09 176602_1	375.00	M5S
10 115481_0	615.00	M5A
10 120075_0	1470.00	M5B

2.2 Green Spaces Dataset

The "Green Spaces" dataset consists of a collection of 3372 geospatial representations of parks, beaches, ravines, etc. in Toronto, and is jointly maintained by the Geospatial Competency Centre and the Parks, Forestry & Recreation division (City of Toronto 2026c). Importantly, this dataset does not include green roofs. The R packages `sf`, `geojsonsf`, and `canadianmaps` were used to parse these representations, record their area, and assign them each to an FSA based on their centroid coordinates.

Table 2: First 4 Rows of Cleaned Green Spaces Data

ID	G_AREA	FSA
1	1222.710	M1V
2	1517.757	M1B
3	180759.352	M2H
4	1252.153	M1W

2.3 Merging Data by FSA

Once parsed, both datasets were joined together by FSA, allowing for effective comparisons of spatial distributions between green spaces and green roofs. 17% (n = 214) of green roof building permits had missing FSAs in the original dataset, and thus were necessarily omitted at this stage. To better measure how "green" FSAs are, controlling of their total area, we defined a new variable, coverage, which is calculated as the proportion of an FSA's total area that is taken up by green roofs (or green spaces). Total area of FSAs was parsed from data available in the `canadianmaps` package, following a similar procedure to "Green Spaces".

Table 3: First 4 Rows of Merged Spatial Summary Data

FSA	geometry	GREEN_ROOF_COVERAGE	GREEN_SPACE_COVERAGE
M1B	list(list(...	0.0001481	0.3201125
M1C	list(list(...	0.0003865	0.1729499
M1E	list(list(...	0.0002761	0.1973799
M1G	list(list(...	0.0003851	0.4185126

3 Results

3.1 Visualizing Current Green Roof Coverage

Figure 1 maps Toronto's green roof coverage by FSA. We immediately observe that coverage is much higher in downtown FSAs when compared to the rest of the City, reaching upwards of 1.5% of the total area covered by green roofs in the FSAs M5B, M5A, and M5T (corresponding to the neighborhoods of Regent Park, Corktown, the Distillery District, Chinatown, and Kensington Market). Notably, the four smaller FSAs that make up the downtown core have significantly lower coverage than their nearest neighbors. Since green roof adoption depends almost entirely on the recency of a development, these observed trends are likely due to disparities in building development rates between (1) the older downtown core, (2) emergent downtown neighborhoods, and (3) slower-paced outer regions of the city.

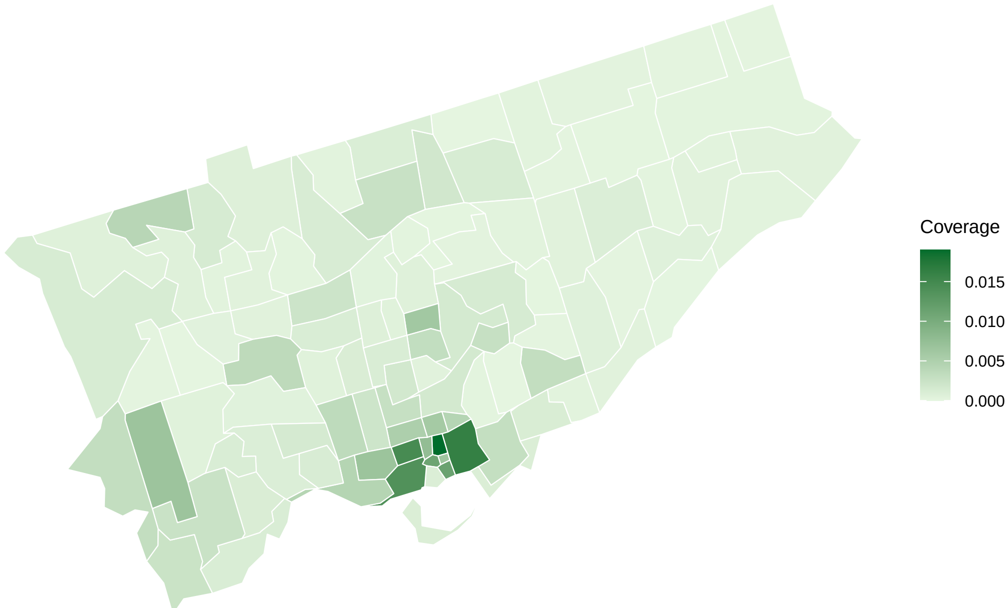

Figure 1: Distribution of Green Roof Coverage in Toronto by FSA.

3.2 Visualizing Current Green Space Coverage

Figure 2 maps green space coverage by FSA. Green spaces are most extensive in Rouge National Park (North-eastern Scarborough ; coverage of up to 94%) and the Harbourfront / Island (coverage of up to 81%) These are expected, as they represent largely forested areas, or areas with high beach coverage. The lowest-coverage FSAs M6B (2%), M4Y (2%), and M5B (3%), which include regions of midtown North York, Downtown Toronto, and the Garden District respectively. Generally, areas with the lowest coverage were scattered across central southern Toronto, while areas with highest coverage could be explained by well-known green spaces. Unlike green roofs, green spaces are present in every FSA, with an average coverage of 17% compared to 0.2%.

3.3 Comparing the Distributions of Green Roofs and Green Spaces

Figure 3 illustrates the observed negative association ($p < 0.05$) between green roof and green space coverage among Toronto's FSAs.

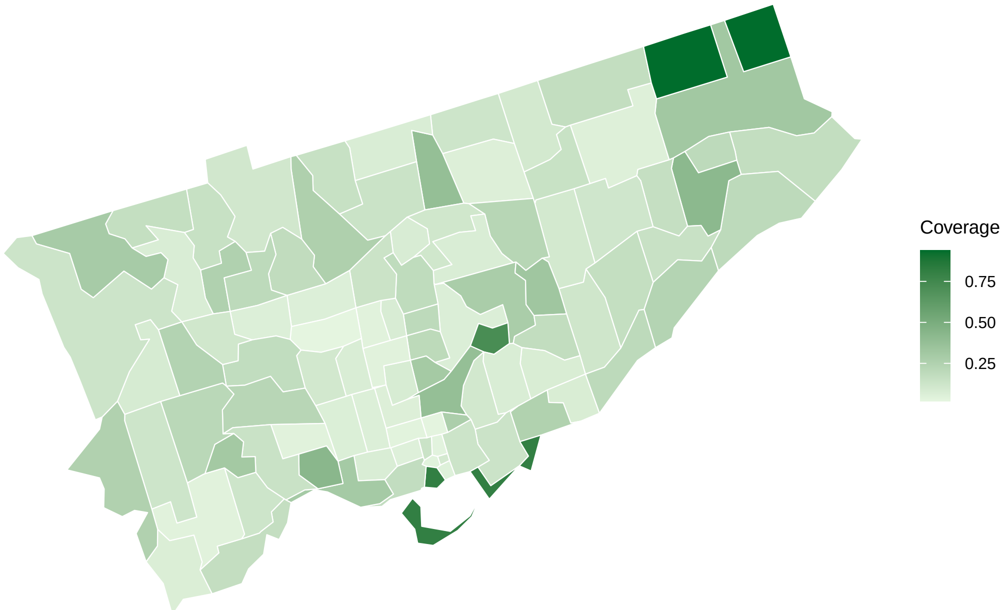

Figure 2: Distribution of Green Spaces in Toronto by FSA.

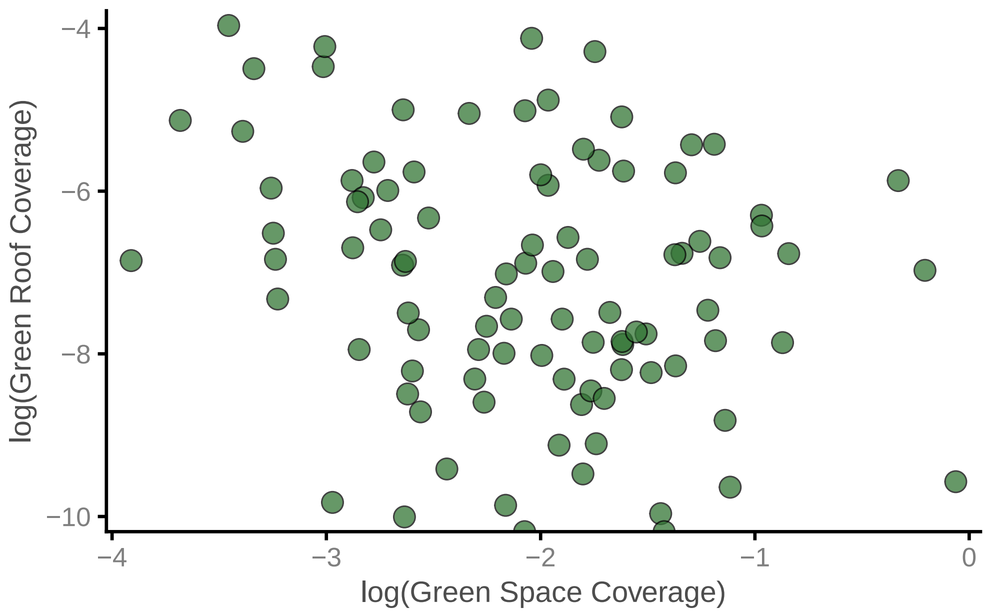

Figure 3: Association between the green roof and green space coverage of Toronto's FSAs (log-log scatterplot)

4 Discussion

In this report, two datasets from Open Data Canada were used to compare the spatial distributions of green spaces and green roofs in Toronto. We found, in accordance with our expectations, that the green space and green roof coverage of FSAs were negatively correlated ; the buildings affected by GRBL and ERIP were built (or are planned to be built) in areas of the city with comparatively less ground-level green spaces, homogenizing the green area in the city. Although not explicitly studied here, we expect that the GRBL was effective in mitigating against the UHI, not only by increasing the absolute amount of green area in Toronto by (at least) 8.5 km^2, but also by reducing spatial inequalities in green coverage.

This observed effect of the GRBL is particularly notable as it does not obviously represent the efforts of targeted ecological policy. The GRBL applied to and was adopted by all structures built over nearly 16 years, meaning that the vast majority of Toronto's green roofs were sited involuntarily ; their spatial distribution and associated benefits are incidental rather than intentional. This is encouraging, and if our observed patterns of development are not unique to Toronto, provides policymakers with evidence that even general-mandate GRBLs can improve the environmental state of their cities.

5 Limitations & Future Directions

FSAs were used as our spatial grain in this study, primarily because of it's accessibility in the dataset. Although this method did capture key green regions of the City (Harbourfront and Rouge National Park), we failed to capture the extensive well-vegetated regions in the Humber and Don valleys, as these cross multiple FSAs. This, paired with a sizeable proportion of unlocated roofs (17%) likely reduced the statistical significance our main result, since these FSAs cannot be assumed to have homogenous coverage. Future studies should aim to precisely geolocate GRBL-associated structures, instead of binning them by FSA. Toronto (see a more fine-grained map of the City's green spaces in (Jung et al. 2025)).

A study by Wong & Lau (Wong and Lau 2013) found that in high-density cities such as Hong-Kong, a large proportion of green roofs experience reduced effectiveness from. over shading. Low roofs in densely developed areas such as downtown Toronto may experience little to no direct sunlight in the winter and spring seasons, representing a serious consideration. A more thorough investigation of green roof's ability to offset the UHI requires looking past the two dimentions analyzed here.

Despite these limitations, our results speak to an important strength of city-wide green roof by-laws: the passive homogenization of greenery. Since Toronto's GRBL was repealed in late 2025, the rate of green roof adoption has quickly plateaued, and green roof area provided by voluntary programs such as the ERIP only represents a small fraction (1.6%) of existing vegetated roof area. As the city continues to develop, the burden of urban cooling is placed entirely on active interventions or voluntary programs. Given their historic non-adoption, repealing the GRBL ensures that the vast majority of future building footprints will be ecologically unproductive, effectively halting the progress made toward a more sustainable urban canopy.

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